

Arunav Kumar

AI/ML Engineer

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SUMMARY

AI/ML Engineer with hands-on experience building and deploying production-grade RAG pipelines, NLP systems, and predictive models. Focused on end-to-end delivery from data pipelines and model training to deployed, usable applications.

WORK EXPERIENCE

Labmentix - AI/ML Engineer Intern

July 2026 – Present

- Performed end-to-end EDA and built a binary classification model to predict Customer Satisfaction (CSAT) on 85,000+ ecommerce support tickets; used Chi-square and Mann-Whitney U tests to statistically confirm response time, support channel, and issue category as key drivers of dissatisfaction.
- Built a custom CNN from scratch (PyTorch) for facial emotion recognition on the FER2013 dataset (35,887 grayscale images, 7 classes), addressing a severe 16:1 class imbalance via weighted cross-entropy loss; improved validation accuracy from 44% to 50% by diagnosing a training plateau and adding learning-rate scheduling.

PROJECTS

DocMind - Retrieval-Augmented Generation (RAG) Document Assistant

[Live](#) | [GitHub](#)

Python | FastAPI | FAISS | Groq LLaMA-3.3-70B | Sentence Transformers

- Built end-to-end RAG pipeline PDF ingestion, FAISS vector search, Groq LLaMA-3.3-70B generation with average query response of ~0.5s, tested on the CUAD legal contracts dataset.
- Designed a modular FastAPI backend with a Streamlit frontend; local Sentence Transformer embeddings (all-MiniLM-L6-v2) eliminate API billing, and a persisted FAISS index supports multi-document querying.

CUAD Contract Analysis Pipeline

[GitHub](#)

Python | FAISS | Sentence Transformers | NLP

- Built a legal-contract processing pipeline for clause extraction and summarization, validated across 50/50 contracts from the CUAD dataset; added a FAISS-based semantic search layer for cross-contract clause lookup.

AI Resume Shortlisting System

[Live](#) | [GitHub](#)

Python | Streamlit | Supabase | Sentence Transformers | spaCy | Claude API

- Built an end-to-end NLP pipeline to parse, rank, and shortlist resumes against job descriptions using Sentence-BERT cosine similarity plus a rule-based confidence router; reduced manual screening effort by 50-60% across 200+ real resumes used in college placement drives.
- Integrated Claude API to generate role-specific interview questions for shortlisted candidates, with bulk upload, an ATS validator, and a Plotly analytics dashboard.

Predictive Maintenance RUL Estimator

[Live](#) | [GitHub](#)

Python | PyTorch | XGBoost | Streamlit | NASA CMAPSS Dataset

- Trained and evaluated a PyTorch LSTM and XGBoost on the NASA CMAPSS turbofan dataset to predict Remaining Useful Life (RUL); LSTM achieved lower RMSE (~14.47 cycles) versus XGBoost (~18 cycles) on the FD001 subset.
- Built an interactive Streamlit dashboard to visualize real-time RUL predictions and flag sensor anomalies.

SKILLS

Programming: Python

Databases: PostgreSQL, Supabase

AI / Machine Learning: NLP, RAG, Sentence Transformers, XGBoost, LSTM, Semantic Search, Groq API, Prompt Engineering

Frameworks & Libraries: PyTorch, Scikit-learn, Hugging Face Transformers, LangChain, spaCy, FAISS

Tools & Platforms: Streamlit, FastAPI, Git, GitHub

Cloud & DevOps: AWS (S3, EC2), Docker

Concepts: Model Deployment, Vector Search, Statistical Testing

EDUCATION

GD Rungta College of Science & Technology - BCA

2023 – 2026

Bhilai, Chhattisgarh | Marks: 68.2% | Coursework: Data Structures, Database Management, AI/ML

CERTIFICATIONS

Generative AI Foundations Certificate — **upGrad x Microsoft** | Microsoft Azure AI Essentials: Workloads and ML on Azure - [LinkedIn Learning](#) | Forage BCG X GenAI Job Simulation